



From solvation to sorption: a comprehensive molecular dynamics study of mefenamic acid conformer behavior in pure scCO₂ and scCO₂/cellobiose systems

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ABSTRACT

Classical and *ab initio* molecular dynamics (MD) simulations were employed to investigate the solvation and sorption behavior of mefenamic acid (MFA) molecules in supercritical CO₂ (scCO₂) and scCO₂/cellobiose systems. Using classical MD with alchemical transformation, the Gibbs free energy of solvation (ΔG_{solv}) for four stable MFA conformers in scCO₂ was calculated along the isochores $\rho = 1.1\rho_c$ and $\rho = 1.69\rho_c$ over a temperature range of 333–483 K. The solvation capacity and conformational selectivity of the medium were found to be maximal at high density and low temperature, with the C2a conformer exhibiting the highest thermodynamic stability. For the first time, *ab initio* MD simulations were performed to gain insight into the conformational preferences, local structure and vibrational density of states spectra of MFA in ternary MFA/scCO₂/cellobiose systems (using cellobiose as a model cellulose fragment), compared to binary MFA/scCO₂ systems. The *ab initio* simulations successfully captured transitions between the type-a and type-b MFA conformers via rotation of the aromatic fragment, providing a direct molecular rationale for why these conformations remain indistinguishable in NMR and IR experiments. The presence of cellobiose stabilizes type-b conformers, restricts the rotational freedom of MFA, and induces the formation of directed hydrogen bonds, primarily via the carboxylic acid group. A direct correlation was established between the hydrogen bonds formation and the red shift of O–H/N–H stretching vibrations in the vibrational density of states spectra. These results elucidate the molecular mechanisms underlying the initial stage of impregnation, linking the thermodynamic propensity of MFA conformers in scCO₂ to their specific interactions with the cellulose surface.

1. Introduction

Supercritical carbon dioxide (scCO₂) has emerged as a unique and environmentally friendly solvent for modern pharmaceutical technologies. Its attractive properties such as tunable solvation power, low viscosity, high diffusivity, and ease of removal, make it exceptionally suitable for advanced processing [1]. These properties are critical for processes such as particle crystallization, the production of solid dispersions, the impregnation of active pharmaceutical ingredients (APIs) into polymeric matrices, and drug micronization [2]. Specifically, scCO₂-based techniques offer a powerful means to modify the bioavailability of poorly soluble drugs, stabilize their metastable polymorphic forms, and engineer controlled-release delivery systems [3].

Aerogels based on nanocrystalline cellulose (NCC) represent promising biocompatible and biodegradable carrier matrices for biomedical applications and drug delivery [4,5]. Their usefulness arises not only from a high specific surface area, a well-developed porous structure, and an abundance of functional groups, which facilitate the effective loading and controlled release of APIs [6–9], but also from other key properties such as high water absorption capacity and favorable textural characteristics. Impregnation using scCO₂ is one of the most promising techniques for loading such materials, supported by its proven efficacy both in the synthesis and subsequent loading of aerogels. This method enables the preservation of the material's porous architecture and facilitates the uniform incorporation of thermolabile compounds without compromising their delicate morphology. For instance, Lopes et al. employed

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scCO₂ to impregnate cellulose aerogels, synthesized from ionic liquids and dried via the same method, with phytol (40 °C, 10 MPa, 24 h) [7]. The achieved loading capacity (up to 50 wt%) exceeded that reported for silica and alginate aerogels [10]. This superior performance is attributed to the cellulose aerogel's greater specific surface area and pore volume, suggesting a surface-driven loading mechanism. Importantly, the morphology of the cellulose fibrils was preserved after treatment, which directly confirms the gentle nature of the scCO₂ method.

Mefenamic acid (MFA), a non-steroidal anti-inflammatory drug classified as Biopharmaceutical Classification System (BCS) Class II (low solubility, high permeability), serves as a representative model API. For such compounds, enhancing solubility and dissolution rate is a critical objective [11]. However, the efficiency of the impregnation process and the properties of the final composite are largely governed by the molecular-scale interactions within the ternary system: drug/aerogel/supercritical fluid. These fundamental mechanisms remain insufficiently understood.

A particular challenge, and a crucial consideration, is accounting for the conformational polymorphism of drug molecules like MFA. As a flexible molecule, MFA can exist in several stable conformations. The four most stable MFA conformers differ in the rotation of the carboxyl group (dihedral angle Φ) and the dimethyl-substituted benzene fragment (dihedral angle Ψ), as illustrated in Fig. 1 [12]. These distinct conformations exhibit significant variations in dipole moment, polarizability, and the spatial arrangement of functional groups. This, in turn, predetermines their differential interactions with both solvent molecules and the functional groups of a carrier matrix. Consequently, the contribution of individual conformers to the overall solubility, the nature of adsorption, and the efficiency of immobilization within the carrier can differ fundamentally. Neglecting this factor in thermodynamic modeling and the analysis of molecular mechanisms can be a significant source of error [13].

Therefore, a comprehensive approach is required for the rational design of supercritical impregnation processes and tailored drug delivery systems. First, it is essential to determine the thermodynamic stability of different MFA conformers within the scCO₂ medium across the operational ranges of temperature and pressure. This step identifies the predominant molecular forms present in the solution prior to contact with the carrier matrix. Second, an atomistic understanding is needed of how these predominant (or alternative) conformers interact with the carrier surface, as this interaction dictates the mechanism and efficiency

of their impregnation.

To address the first objective, this work involved calculating the Gibbs free energy of solvation (ΔG_{solv}) for the principal MFA conformers in scCO₂. This calculation provides a quantitative assessment of their relative thermodynamic stability and the conformational selectivity of the supercritical medium.

To address the second objective, an atomistic investigation of molecular interactions in a confined environment, we employed *ab initio* MD. This computationally intensive method was chosen due to the necessity of modeling electronic-structure-dependent properties that are inaccessible to classical simulations using parameterized force fields with fixed atomic charges. *Ab initio* MD provides a direct description of chemical bonding, polarization, and their temporal evolution, which is critical for accurately modeling the formation and spectral signatures of hydrogen bonds, the vibrational density of states (VDOS), and ultrafast conformational transitions. These capabilities enable a direct correlation between intermolecular interactions, VDOS, and molecular dynamics, which is essential for interpreting experimental data. Given the high computational cost of *ab initio* MD, cellobiose (Fig. 1), a glucose dimer containing the key functional groups of cellulose, was used as a chemically relevant and computationally tractable model for the NCC surface [14,15]. This approach provides a rigorous first-principles foundation for analyzing the structural, vibrational, and conformational changes of MFA at the cellulose interface.

While a substantial body of work exists on modeling pure scCO₂ [16,17], solutions therein [18,19], and multiphase systems [20–23], comprehensive studies that combine thermodynamic analysis of the API's conformational state with detailed *ab initio* modeling of ternary systems (API/scCO₂/carrier) [24], remain exceedingly rare. We therefore expect that the findings of the present study will contribute to filling this knowledge gap and provide a robust molecular framework for the rational design of supercritical impregnation processes.

2. Simulation details

2.1. Solvation free energy calculation via classical molecular dynamics

The Gibbs free energy of solvation (ΔG_{solv}) for MFA conformers was calculated using the Bennett Acceptance Ratio (BAR) method, implemented in the Gromacs-5.0.7 software package [25]. The calculation followed a standard alchemical transformation protocol, where the

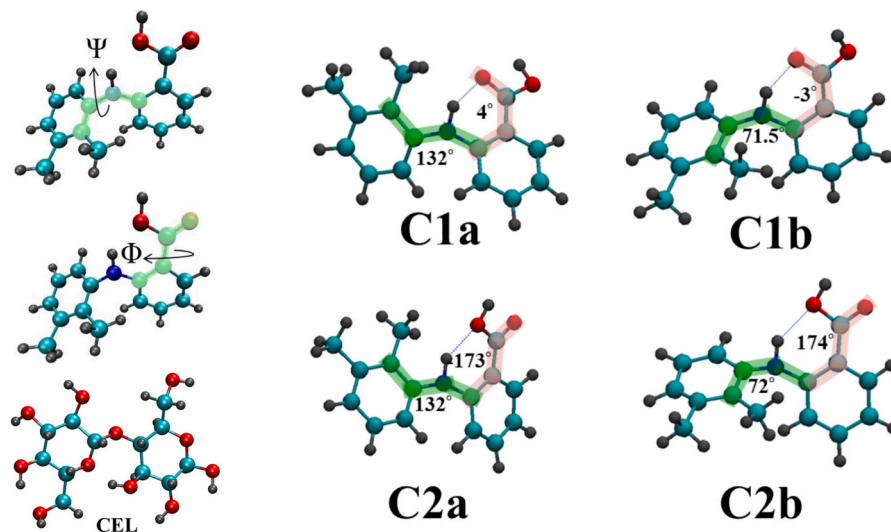


Fig. 1. Molecular structure of mefenamic acid highlighting the key torsional degrees of freedom. The rotation of the dihedral angles Φ and Ψ , defined by the green atoms, enables interconversion between the distinct MFA conformers. Schematic representations of cellobiose (CEL), a glucose dimer containing the key functional groups of cellulose, and the four identified MFA conformers (C1a, C1b, C2a, C2b) are shown. The dihedral angles Φ and Ψ are taken from [12] for B3LYP–GD3 functional. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

interactions of the MFA molecule with its environment were gradually turned on using a coupling parameter λ , which varied from 0 (no solute) to 1 (fully interacting solute). To ensure accuracy and a smooth transition, 15 independent simulations were performed in the *NPT*-ensemble for the following λ values: 0, 0.05, 0.1, 0.2, 0.3, 0.4, 0.45, 0.5, 0.55, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0. Simulations employed a stochastic integrator with a 2 fs timestep. Pressure was controlled using the Parrinello-Rahman barostat [26] with a time constant of 1.0 ps and a compressibility of $7.8 \times 10^{-4} \text{ bar}^{-1}$. Long-range electrostatic interactions were calculated using the Particle Mesh Ewald (PME) method with a cutoff of 1.2 nm, while van der Waals interactions were treated using a Lennard-Jones potential with a 1.2 nm cutoff. For small λ values, a soft-core potential with parameters $\text{sc-alpha} = 1.0$ and $\text{sc-sigma} = 0.3 \text{ nm}$ was applied. Long-range dispersion corrections for energy and pressure were employed. All bonds involving hydrogen atoms were constrained using the LINCS algorithm.

For each λ value, the system underwent energy minimization, followed by a 0.5 ns equilibration in the *NPT*-ensemble, and a subsequent 10 ns production run. The adequacy of the 0.5 ns equilibration time was confirmed by monitoring the convergence of temperature and pressure in extended test simulations (Supplementary Material Fig. S1). The energy differences required for Bennett's method analysis were saved every 0.1 ps. Carbon dioxide was represented by the Aime et al. model [28]. For all atoms of the MFA molecule, the Lennard-Jones potential parameters (ϵ and σ) and partial atomic charges were obtained automatically using the LigParGen server [29–31], which employs the OPLS-AA force field [27]. This approach is standard for organic molecules and ensures compatibility between the solute and solvent parameters. The C1a conformer was selected as the initial molecular geometry. Using the Open Babel program [32], we performed rotations of the carboxyl group and the aromatic ring around their respective bonds to generate a set of conformers (using the dihedral angles corresponding to the B3LYP-GD3 data from [12]) for subsequent analysis. To prevent undesired conformational transitions during the alchemical transformation, the rotational barriers for key dihedral angles defining the conformations were restrained using harmonic potentials with increased force constants.

We performed calculations along two isochores, $\rho = 1.1\rho_c$ and $\rho = 1.69\rho_c$ (where $\rho_c = 10.6249 \text{ mol/L}$ is the critical density of CO_2), at temperatures of 333 K, 403 K, and 483 K. The corresponding pressures were taken from the NIST Chemistry WebBook [33] and were 130.91 bar ($T = 333 \text{ K}$), 274.3 bar ($T = 403 \text{ K}$), 437.22 bar ($T = 483 \text{ K}$) for $\rho = 1.1\rho_c$, and 252.9 bar ($T = 333 \text{ K}$), 595.76 bar ($T = 403 \text{ K}$), 981.49 bar ($T = 483 \text{ K}$) for $\rho = 1.69\rho_c$. These two isochores were selected for investigation based on prior experimental and MD research by our group on related MFA/ scCO_2 systems [12,34]. These densities correspond to significant regions of the supercritical phase diagram where specific solute-solvent interactions and the conformational behavior of MFA have been established, thereby anchoring our simulations in experimentally relevant conditions. All simulations were performed in a cubic box with periodic boundary conditions containing 1 MFA molecule and 512 CO_2 molecules.

2.2. *Ab initio* molecular dynamics simulation

Ab initio MD simulations were performed using the CP2K software package, version 2023.1 [35]. The electronic structure was calculated within the density functional theory (DFT) framework using the Gaussian and Plane Waves (GPW) method. The PBE functional was employed as the exchange-correlation functional. For all atoms, double-zeta basis sets with polarization functions (DZVP-MOLOPT-GTH) and norm-conserving Goedecker-Teter-Hutter (GTH-PBE) pseudopotentials were used [36,37]. The plane wave energy cutoff was set to 400 Ry. To account for dispersion interactions, an empirical Grimme-type DFT-D3 correction with a cutoff radius of 7 Å was applied [38,39].

Simulations were conducted in the *NVT*-ensemble within rectangular or cubic simulation boxes with periodic boundary conditions.

Temperature was maintained using a Nosé-Hoover thermostat with a chain length of 3, a Yoshida parameter of 3, a multi-level integration scheme (MTS = 2), and a time constant of 1000 fs. The total production run duration was 24 ps with an integration timestep of 0.5 fs. Atomic coordinates were saved at each step, while velocities and forces were recorded every 5 fs. To confirm the thermodynamic stability of our simulations, the time evolution of the system temperature for all production runs is provided in Fig. S2 (Supplementary Material), showing stable convergence to the target values.

For the eight distinct systems studied (compositions and parameters are listed in Table 1), initial configurations were generated using the Packmol program [40]. The composition was chosen to ensure the full solvation of the cellobiose and MFA molecules within a computationally manageable box size, focusing on the fundamental interactions rather than replicating a specific bulk experimental condition. For each system, energy minimization via the steepest descent method and a 100 ps *NVT*-ensemble equilibration were performed in Gromacs-5.0.7 [25] using the potential models for CO_2 and MFA described in Section 2.1 and for cellobiose (CEL) OPLS-AA force field [27] generated using the LigParGen server [29–31]. This equilibration time was intentionally kept short to avoid significant deviation from the initial molecular packing. The adequacy of this timeframe was confirmed by monitoring the temperature stability, which showed no drift beyond 100 ps in extended test runs (Supplementary Material Fig. S1). Critically, and in contrast to the restrained simulations used for ΔG_{sol} computations, no dihedral angles were fixed here. This allowed the system to explore a broader conformational space, which is essential for obtaining representative starting structures for the subsequent *ab initio* MD sampling. The resulting final configurations served as the starting points for subsequent *ab initio* MD simulations. Final snapshots of the *ab initio* MD simulations are presented in Fig. 2.

Radial distribution functions, bond lengths, angles, and average numbers of hydrogen bonds (based on the geometric criteria: $r(\text{O} \cdots \text{O}) \leq 0.35 \text{ nm}$, $\angle \text{H}-\text{O} \cdots \text{O} \leq 30^\circ$) were calculated using the VMD software [41]. Power spectra (VDOS) were computed using the TRAVIS program [42,43].

It is important to note that the temperatures for the classical (ΔG_{sol})

Table 1

Parameters of the simulated systems: composition (number of molecules, N), temperature (T), simulation box dimensions (L), density (ρ), and CO_2 concentration (C).

System	N (MFA)	N (CO_2)	N (CEL)	T, K	$L_x \times L_y \times L_z$, nm ³	ρ , g/cm ³	C(CO_2), mol/L
1	1	64	0	328	$2.1 \times 2.1 \times 2.1$	0.548	11.4755
2	1	64	0	433	$2.1 \times 2.1 \times 2.1$	0.548	11.4755
3	1	64	0	473	$2.1 \times 2.1 \times 2.1$	0.548	11.4755
4	1	40	4	433	$1.5 \times 1.5 \times 1.9$	1.309	15.5372
5	1	40	4	433	$1.5 \times 1.5 \times 1.9$	1.309	15.5372
6	1	40	4	433	$1.5 \times 1.5 \times 1.9$	1.309	15.5372
7	1	62	2	433	$1.5 \times 1.5 \times 1.5$	1.798	30.5047
8	1	32	4	433	$1.5 \times 1.5 \times 1.5$	1.485	15.7443

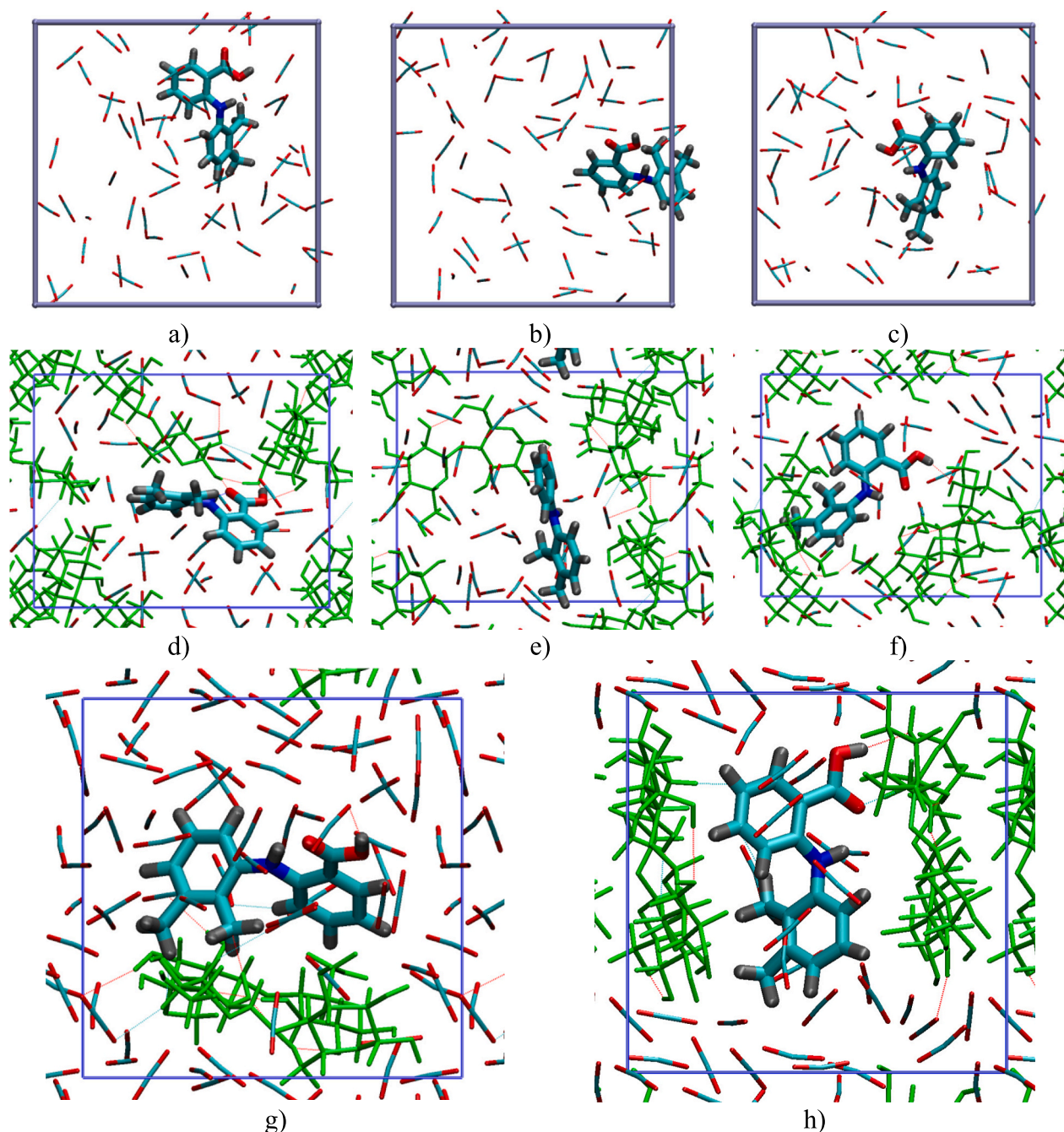


Fig. 2. Final simulation snapshots of all systems: (a) System 1 (MFA/64CO₂ at 328 K), (b) System 2 (MFA/64CO₂ at 433 K), (c) System 3 (MFA/64CO₂ at 473 K), (d) System 4 (MFA/4CEL/40CO₂ at 433 K), (e) System 5 (MFA/4CEL/40CO₂ at 433 K), (f) System 6 (MFA/4CEL/40CO₂ at 433 K), (g) System 7 (MFA/2CEL/62CO₂ at 433 K), and (h) System 8 (MFA/4CEL/32CO₂ at 433 K).

and *ab initio* MD simulations were chosen independently, based on the optimal conditions and specific objectives of each methodological approach. The classical MD calculations were performed at 333, 403, and 483 K to efficiently map the free energy landscape along the selected isochores and capture the crossover between enthalpic and entropic regimes. In contrast, the *ab initio* MD simulations were conducted at 328, 433, and 473 K. This choice was motivated by practical computational constraints and the goal of modeling conditions directly comparable to previous experimental and simulation studies of scCO₂ structure and dynamics [12,34]. Despite this difference, the selected temperatures are sufficiently close (within ~5–10 K) to allow for a meaningful comparative analysis. Both sets of temperatures lie within the same operational window for scCO₂-based processes and reveal consistent trends related

to the effects of density and temperature.

The present *ab initio* MD study does not aim to investigate the preferential formation of specific MFA conformers in scCO₂ across different temperatures or under confinement. The primary reason for this is the inherent limitation of the simulation time: while the 24 ps trajectory is characteristic of resource-intensive *ab initio* calculations, it is generally insufficient for exhaustive sampling of the conformational space or for capturing transitions that may require hundreds of nanoseconds to microseconds to occur. Instead, this work characterizes the behavior of pre-selected MFA conformers under the studied conditions and captures the conformational rearrangements that are accessible within this representative, yet limited, timeframe, specifically the C1a⇌C1b and C2a⇌C2b equilibria.

3. Results and discussion

3.1. Gibbs free energy of solvation for MFA conformers

Along both isochores, $\rho = 1.1\rho_c$ and $\rho = 1.69\rho_c$, a pronounced temperature dependence of ΔG_{solv} is observed (Fig. 3): as temperature increases, the value of ΔG_{solv} becomes less negative. This increase in ΔG_{solv} for the MFA conformers with rising temperature is consistent with the classical behavior of processes dominated by enthalpic contributions. Solvation can be viewed as a competition between energetically favorable solute-solvent interactions and the entropically unfavorable ordering of solvent molecules around the solute. Upon heating, the enthalpic contribution (ΔH) weakens due to the disruption and destabilization of intermolecular interactions (van der Waals forces, dipole-quadrupole interactions, and weak hydrogen bonds) caused by intensified thermal motion. Concurrently, the entropic contribution ($-T\Delta S$) becomes more positive, as a more disordered state becomes thermodynamically favorable for the system. The increasing influence of entropy leads to the destabilization of the structured solvation shell, which is clearly evident from the behavior of the center-of-mass radial distribution function (RDF) for MFA- CO_2 (Fig. 4).

Across the entire investigated temperature range, a pronounced dependence of the solvation free energy on scCO_2 density is observed. Upon transitioning to the isochore $\rho = 1.69\rho_c$, the ΔG_{solv} values become significantly more negative compared to those at $\rho = 1.1\rho_c$ for temperatures of 333 K and 403 K. Increasing the solvent density leads to a higher number of solvent molecules per unit volume, thereby enhancing solute-solvent intermolecular interactions. This change in the microenvironment results in a substantial increase in the solvation capacity of the medium. For instance, the number of contacts between any atom of the MFA conformers (calculated for C1a and C2b) and carbon atoms of CO_2 molecules within a 0.6 nm radius increases from 21 ± 4 to 32 ± 3 as density rises from $1.1\rho_c$ to $1.69\rho_c$ at 333 K. Thus, increasing scCO_2 density is a key factor enhancing solvation and, consequently, thermodynamically stabilizing all MFA conformers in the studied systems.

However, at the elevated temperature of 483 K, an inverse trend is observed in the Gibbs free energy of solvation with increasing scCO_2 density. Unlike at 333 K and 403 K, ΔG_{solv} becomes less negative for all conformers when transitioning from the $1.1\rho_c$ to the $1.69\rho_c$ isochore. This suggests that at 333 K and 403 K, the dominant contribution to ΔG_{solv} is enthalpic: an increased number of CO_2 molecules in the MFA solvation shell leads to stronger, energetically favorable interactions, while the entropic penalty from solvent ordering at moderate temperatures remains relatively minor. In contrast, at 483 K, the entropic contribution to the free energy increases sharply. The entropy loss

caused by structuring a dense solvation shell results in a positive (unfavorable) $-T\Delta S$ term, which outweighs the enthalpic gain.

An analysis of the calculated ΔG_{solv} values allows us to rank the MFA conformers in order of decreasing solvation stability in scCO_2 across the studied state parameters: $\text{C2a} > \text{C2b} \geq \text{C1a} \geq \text{C1b}$. The conformer C2a exhibits the most negative ΔG_{solv} values under all temperature and density combinations, indicating an optimal spatial arrangement of its functional groups for efficient interaction with solvent molecules and the formation of a thermodynamically stable solvation complex. C2b ranks second, showing less favorable solvation than C2a but a significantly greater solvation propensity in scCO_2 compared to the C1a and C1b conformers. The latter pair is characterized by the highest ΔG_{solv} values, with C1b being the least energetically favorable for solvation under all investigated conditions.

It is important to emphasize that the solvation capacity of scCO_2 toward different conformers, as discussed above, strongly depends on thermodynamic parameters. The difference in ΔG_{solv} reaches a maximum under high-density, low-temperature conditions, amounting to approximately 3.7 kJ/mol between conformers C2a and C1b at 333 K and a density of $1.69\rho_c$. Conversely, at low density and high temperature, these differences are significantly reduced, decreasing to as low as 0.74 kJ/mol. This observation indicates that under high-density, low-temperature conditions, the solvation environment exhibits heightened selectivity and an enhanced ability to differentiate between the structural variations of the conformers. In contrast, as temperature increases, entropic effects diminish these distinctions.

3.2. Influence of temperature and cellobiose presence on the conformational behavior of MFA in scCO_2

The thermodynamic analysis conducted in the previous section revealed the relative stability of different MFA conformers in the bulk scCO_2 phase and allowed them to be ranked according to their solvation capacity. However, to understand the atomistic mechanisms underlying the impregnation process, it is essential to investigate the populations of conformers under conditions closely approximating the actual process – that is, in close proximity to the carrier surface. Furthermore, free energy calculations do not provide information on the dynamics of conformational transitions, the local structure of the solvation shell, or the nature of specific interactions with cellulose's functional groups. To address these aspects and obtain a detailed molecular picture of the initial sorption stage, *ab initio* MD simulations of ternary MFA/ scCO_2 /cellobiose systems, as well as control binary MFA/ scCO_2 systems at various temperatures, were performed.

To analyze the conformational behavior of MFA in pure scCO_2 at various temperatures and upon interaction with cellobiose, the distributions of the dihedral angles Φ and Ψ were calculated over the entire production simulation trajectory, along with their corresponding probability distributions (Figs. 5 and 6). On the graphs, multicolored squares indicate the angle values corresponding to the most stable MFA conformers as determined by quantum chemical methods [12].

As shown in Fig. 5, the MFA molecule, initially adopting a conformation close to C2a, explores a wide range of Φ and Ψ values during the simulation. At 328 K, the most probable values correspond to those characteristic of the C2a conformer. As the temperature increases, the angular dispersion broadens, and the probability of the dimethyl-substituted aromatic group transitioning into the C2b conformation rises. Specifically, the distribution of the Ψ angle, which is unimodal at 328 K, becomes significantly broader at 473 K, exhibiting a pronounced maximum around 75° . Additionally, the mobility of the carboxyl group increases, with the distributions of the Φ angle becoming wider and their amplitude decreasing.

Systems 4–6, which have identical compositions and were simulated at the same temperature (433 K), differed in the initial conformation of MFA. Although the simulation for System 4 began with the C1a conformation, frequent conformational transitions were observed

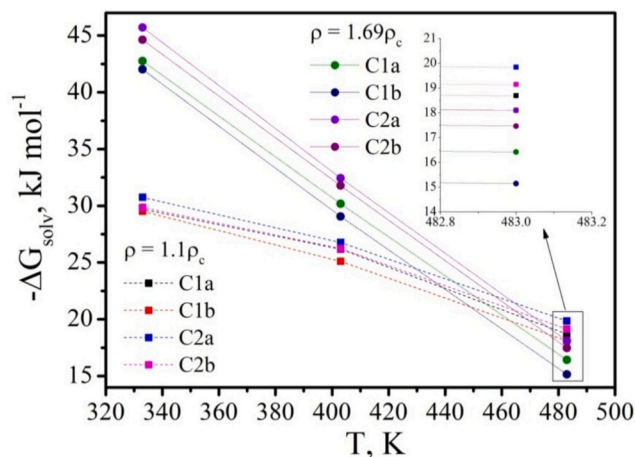


Fig. 3. Temperature dependence of the Gibbs free energy of solvation (ΔG_{solv} , kJ/mol) for MFA conformers along the isochores $\rho = 1.1\rho_c$ and $\rho = 1.69\rho_c$.

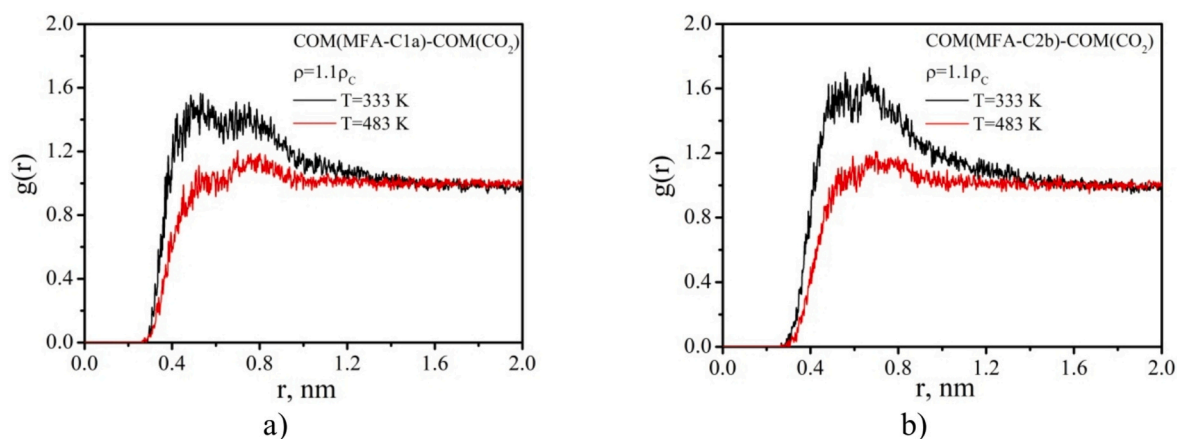


Fig. 4. Center-of-mass (COM) radial distribution functions of mefenamic acid conformers C1a and C2b relative to carbon dioxide molecules along the isochore $\rho = 1.1\rho_c$ at two temperatures: 333 K and 483 K.

during the simulation, primarily between the C1a and C1b states, with the C1b conformation being the most probable. A similar pattern was observed for System 6: the dihedral angle distributions also indicate the dominance of the C1b conformation (Fig. 6c). However, it should be noted that the Ψ angle distribution in System 6 is more symmetrical compared to that in System 4.

In the presence of cellobiose, the distribution of the angle Φ , which describes the rotation of the carboxyl group, becomes narrower than in System 2, where MFA is in pure $scCO_2$. This indicates that the interaction between MFA and cellobiose stabilizes specific conformations, restricting the rotational freedom of both the carboxyl group and the aromatic fragment compared to the pure $scCO_2$ environment. Furthermore, the presence of cellobiose increases the propensity of MFA to adopt a type-b conformer. This is reflected in a bimodal distribution of the angle Ψ , with two distinct maxima of approximately equal probability at $\Psi \approx 75^\circ$ and 105° .

In systems 7 and 8, MFA initially adopted a conformation close to C1b. During MD simulations, the distributions of the torsional angles Φ and Ψ were concentrated in regions characteristic of this conformation (Fig. 7). However, the group dynamics in the two systems differed significantly. Slightly narrower and more compact angle distributions were observed in System 7, where the carboxyl group is oriented toward the bulk $scCO_2$ phase. In System 8, despite the formation of hydrogen bonds with cellobiose (as shown below), the distributions were somewhat broader. This difference can be explained by the distinct nature of mobility constraints. In System 7, the relatively homogeneous environment does not impose strict steric or electrostatic restrictions on specific angles. The carboxyl group readily finds and stably maintains the most energetically favorable conformation, leading to a narrow distribution. In System 8, hydrogen bonds with various sites on cellobiose create competing steric and electrostatic constraints. The carboxyl group “switches” between several metastable states corresponding to different bond acceptors and donors, resulting in Brownian-like motion within a broader range of Φ and Ψ angles. Thus, the formation of specific hydrogen bonds with cellobiose in System 8 does not lock the conformation; on the contrary, it creates multiple near-degenerate energetic states, increasing conformational heterogeneity.

The thermodynamic analysis in the bulk $scCO_2$ phase (Section 3.1) clearly demonstrates that type-a conformers (C2a, C1a) have a more negative ΔG_{solv} and are therefore thermodynamically more favorable in the pure solvent environment than their type-b counterparts. However, *ab initio* MD simulations in the presence of cellobiose (Systems 4–8) reveal a different scenario: the dihedral angle distributions (Figs. 6 and 7) show a clear preference for MFA to adopt type-b conformers (C1b, C2b), despite their lower stability in the bulk phase.

This result indicates that a different stabilization factor begins to

dominate near the cellobiose molecules, outweighing the solvation preferences of the bulk $scCO_2$ phase. It can be hypothesized that the spatial arrangement and accessibility of functional groups in type-b conformers are more optimal for forming specific interactions (particularly hydrogen bonds) with the hydroxyl groups of cellobiose. Thus, the carrier surface does not merely passively accept conformers “prepared” by the solvent. Instead, it actively selects those molecular forms that best match its own structural and chemical features, providing an important insight into the impregnation mechanism.

The conformational states a and b of MFA appear merged and inseparable in NMR and IR spectra, both in pure $scCO_2$ and under aerogel confinement in $scCO_2$ [12,44,45]. Using *ab initio* MD, we demonstrate why this spectral resolution is impossible: the $a \rightleftharpoons b$ interconversion is an ultrafast process. Our 24-ps trajectories directly captured these rapid transitions, explaining the time-averaged, featureless signal observed in experiment.

3.3. Analysis of intermolecular interactions via VDOS spectra and RDFs

The vibrational density of states spectra for the complex systems as a whole, as well as for their individual components, were calculated from the *ab initio* MD trajectories using the TRAVIS program [42,43] (Fig. S3).

VDOS spectra of $scCO_2$ (Fig. 8), derived from *ab initio* MD trajectories, display characteristic features that reflect the influence of the dense fluid environment on intramolecular modes. Analysis of the VDOS spectrum (Fig. 8) reveals four distinct absorption regions. Interpreting these features offers insights into the nature of intermolecular interactions within the system.

The most intense low-frequency band is observed at approximately $621\text{--}635\text{ cm}^{-1}$, accompanied by a small shoulder in the $706\text{--}737\text{ cm}^{-1}$ region. This band corresponds to the bending mode (ν_2) of CO_2 , which for an isolated linear molecule appears near 667 cm^{-1} . The significant red shift of $\sim 32\text{--}46\text{ cm}^{-1}$ is a direct consequence of steric hindrance imposed by the dense molecular environment of the supercritical fluid. To execute the bending vibration, the molecule must overcome a potential barrier associated with the rearrangement of its nearest neighbors, which effectively increases the inertial resistance and lowers the vibrational frequency. The shoulder at $706\text{--}737\text{ cm}^{-1}$ can be interpreted as a signature of the deformation of CO_2 from a strictly linear geometry in the dense fluid phase. An alternative explanation involves orientational ordering, where prevalent T-shaped or other non-collinear configurations between CO_2 molecules could split or shift components of the bending mode due to symmetry breaking, without requiring a permanent bend in the isolated molecule. These observations are consistent with earlier *ab initio* simulations of $scCO_2$ [16].

In the mid-frequency region, a doublet is observed, consisting of

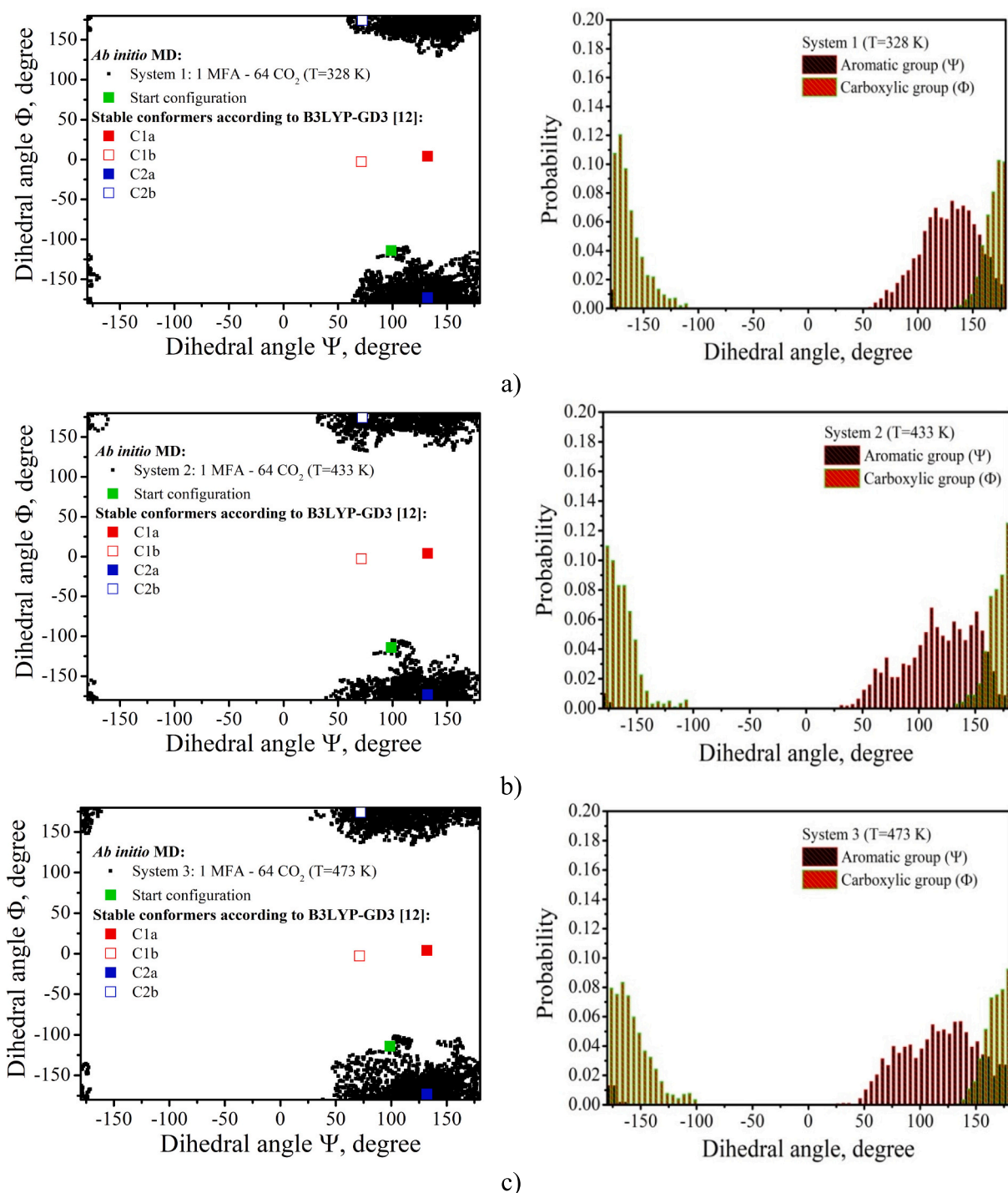


Fig. 5. Distribution of dihedral angles Φ and Ψ of MFA molecule in scCO_2 : Systems 1–3.

bands at $\sim 1214\text{--}1248\text{ cm}^{-1}$ and $\sim 1349\text{--}1358\text{ cm}^{-1}$. This pair directly manifests the resonance Fermi interaction between the first overtone of the bending vibration ($2\nu_2$) and the fundamental symmetric stretching vibration (ν_1) [46]. Under supercritical conditions, the initially close-in-energy $2\nu_2$ and ν_1 levels undergo strong interaction, leading to their repulsion and the formation of two hybridized states. The band at $1214\text{--}1248\text{ cm}^{-1}$ predominantly retains the character of the bending overtone ($2\nu_2$), while the band at $\sim 1349\text{--}1358\text{ cm}^{-1}$ mainly exhibits the properties of the symmetric stretching vibration ν_1 . Experimentally, these bands are observed at 1281 and 1387 cm^{-1} , respectively, in the supercritical state [47,48], and at 1388 cm^{-1} for the isolated molecule.

The intensification of this resonance in the supercritical state compared to the gas phase results from shifts in the vibrational levels induced by intermolecular interactions [49].

The asymmetric stretching vibration ν_3 of the CO_2 molecule, appearing as an intense band in the range of $\sim 2344\text{--}2363\text{ cm}^{-1}$ (depending on the system), serves as a sensitive probe of its microenvironment. In the gas phase, this vibration is observed as a narrow line near 2349 cm^{-1} . However, significant changes in its spectral parameters occur in the condensed MFA/ scCO_2 and MFA/ scCO_2 /cellobiose systems. The position of the band maximum varies significantly with the system composition, covering a range from 2344 cm^{-1} to 2363 cm^{-1} . This

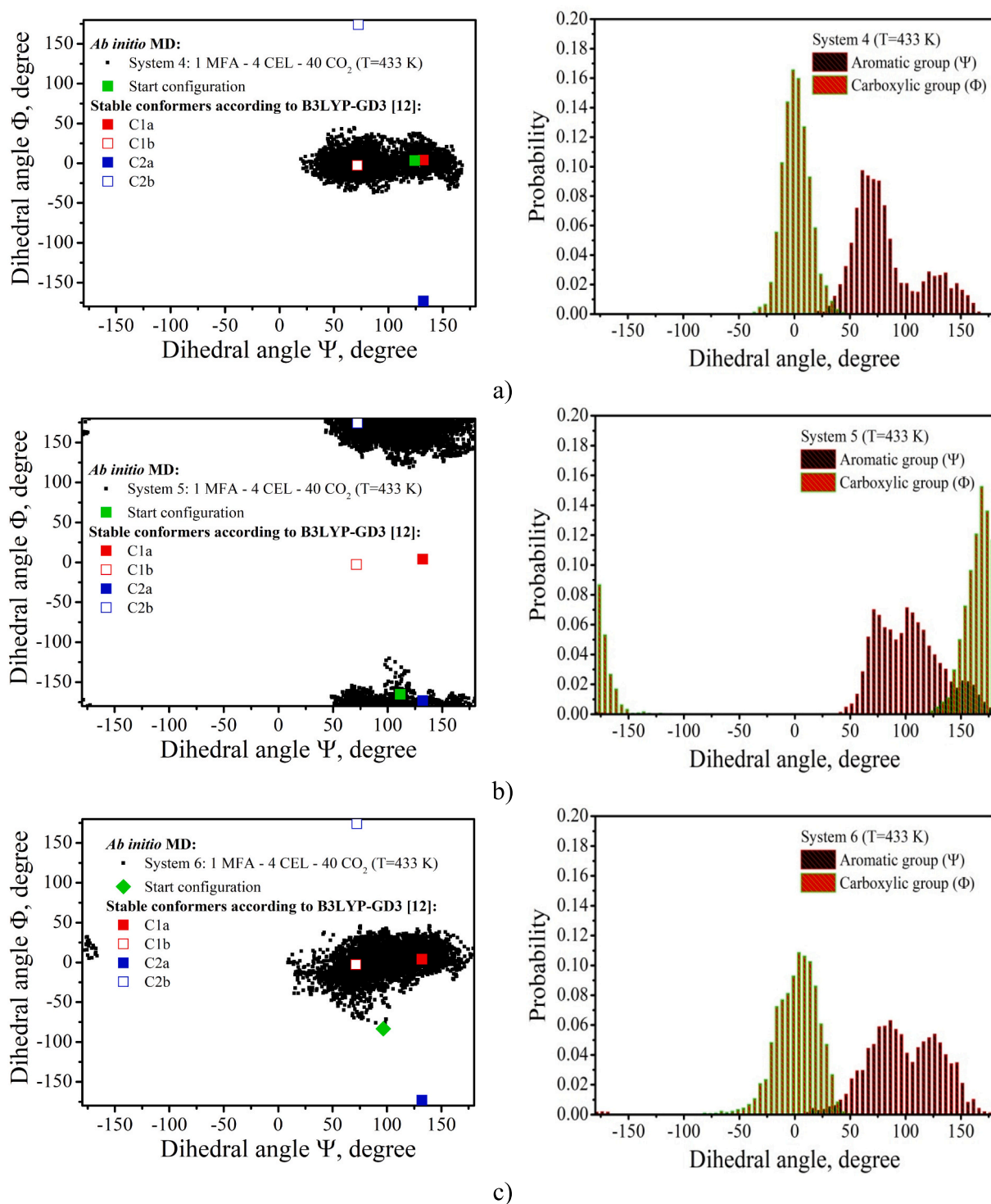


Fig. 6. Distribution of dihedral angles Φ and Ψ of MFA molecule in scCO_2 /cellobiose: Systems 4–6.

broad shift demonstrates the high sensitivity of this vibration to changes in the intermolecular interactions, including the potential formation of solvation complexes with solution components.

Thus, the calculated VDOS spectrum of scCO_2 provides direct evidence of the complex and selective influence of the supercritical environment on different molecular degrees of freedom. The sensitivity of the asymmetric stretching vibration ν_3 to the system's composition indicates the formation of specific intermolecular complexes. The bending mode ν_2 , in turn, exhibits a pronounced low-frequency shift, reflecting a global weakening of bonds due to the influence of the polarizable

environment and steric constraints. The most significant changes occur in the region of symmetric stretching vibrations, associated with the intensification of the Fermi resonance, highlighting the medium's ability to indirectly influence intramolecular interactions. These results unequivocally demonstrate that the supercritical fluid acts not as an inert medium but as an active participant in the processes occurring within it.

The slight increase in band intensity and broadening in the calculated VDOS spectra of CO_2 with rising temperature at constant density (Fig. 8a) arise from weakened intermolecular interactions and a transition toward more gas-like vibrational dynamics. The RDFs for CO_2

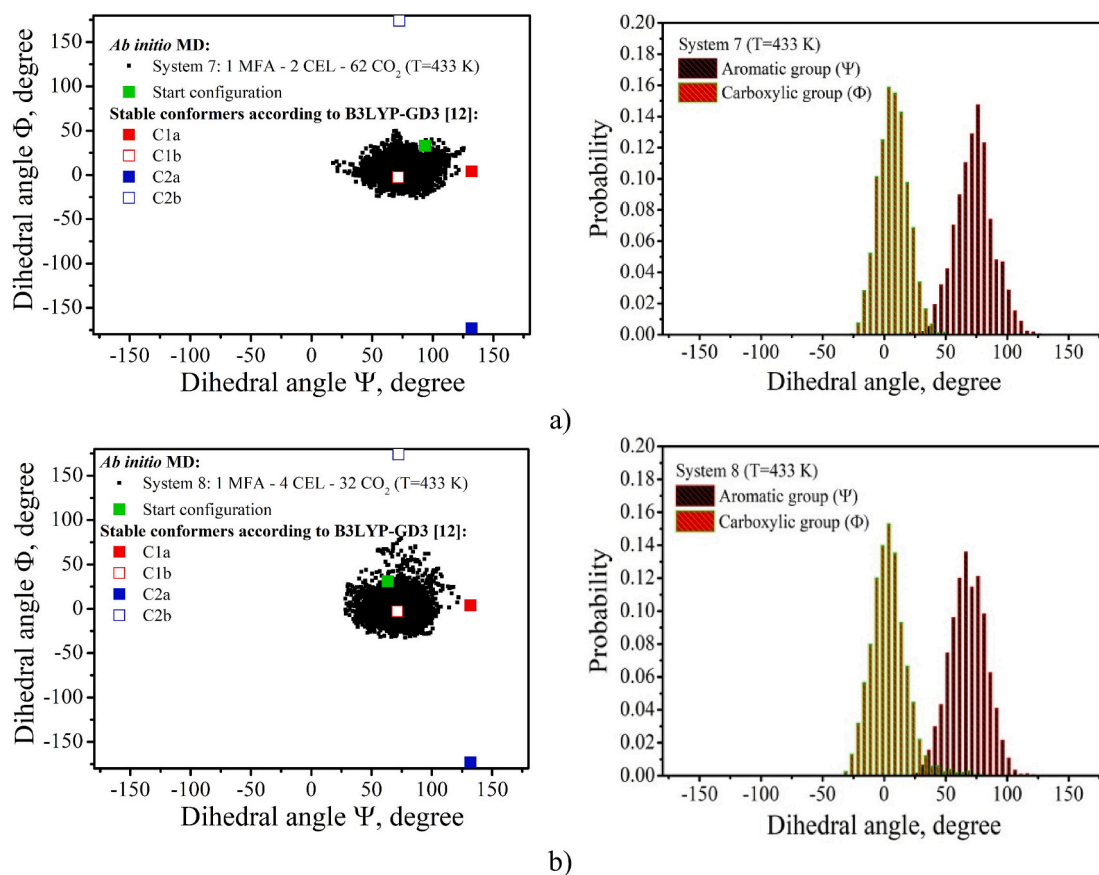


Fig. 7. Distribution of dihedral angles Φ and Ψ of MFA molecule in scCO_2 /cellobiose: Systems 7–8.

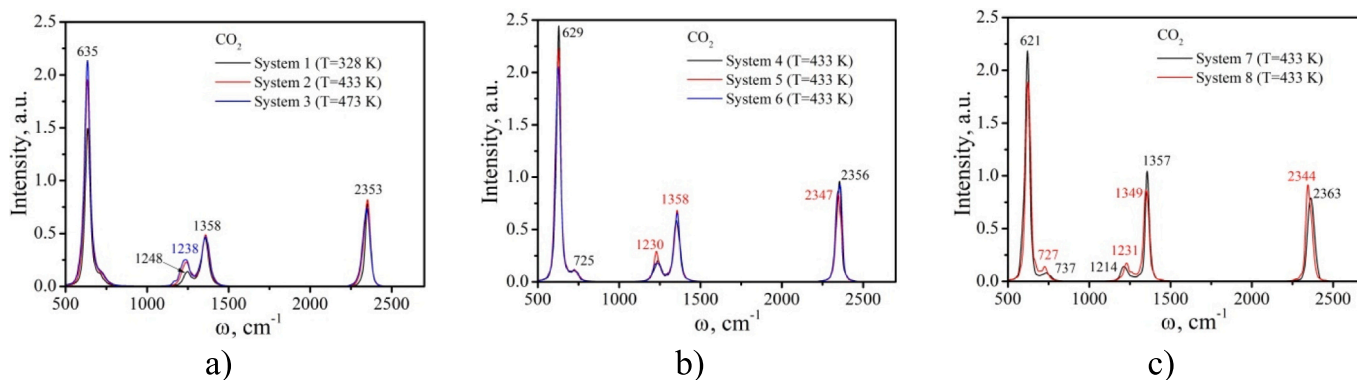


Fig. 8. Vibrational density of states spectra of carbon dioxide (a) in Systems 1–3 (1 MFA – 64 CO_2), (b) in Systems 4–6 (1 MFA – 4 CEL – 40 CO_2), (c) in System 7 (1 MFA – 2 CEL – 62 CO_2) and System 8 (1 MFA – 4 CEL – 32 CO_2).

atoms relative to each other (Fig. 9) show that, for Systems 1–3, the height of the first peaks significantly decreases with increasing temperature, reflecting a structural change toward a less ordered local environment. This reduction in local order frees the intramolecular vibrations, decreasing the damping effect induced by the surrounding environment. This effect outweighs potential thermal line broadening and results in an increased amplitude of vibrational modes, as recorded in the VDOS spectrum. Furthermore, the temperature rise enhances anharmonic effects and alters the conditions of the Fermi resonance, further contributing to the observed changes in the intensities of the Fermi doublet. Therefore, the temperature dependence of band intensity serves as a sensitive indicator of the restructuring of intermolecular interactions within scCO_2 , even under conditions of constant macroscopic

density.

Regarding the structure of scCO_2 , we observe that for systems with low to moderate density (Systems 1–6), the $\text{C}(\text{CO}_2)\text{-O}(\text{CO}_2)$ RDF exhibits a characteristic pre-peak in the range of 0.30–0.35 nm (Fig. 9b). This feature is accurately reproduced by our model and is consistent with literature reports describing the predominant distorted T-shaped orientation of nearest-neighbor CO_2 molecules in the supercritical state [17,50]. The T-shaped arrangement of CO_2 molecules is attributed to Lewis acid–Lewis base (LA–LB) donor–acceptor interactions. The possibility of such interactions arises from the presence of lone electron pairs on oxygen atoms and a partial positive charge on the electron-deficient carbon atom in the CO_2 molecule.

A significant structural change is observed in System 7, which has the

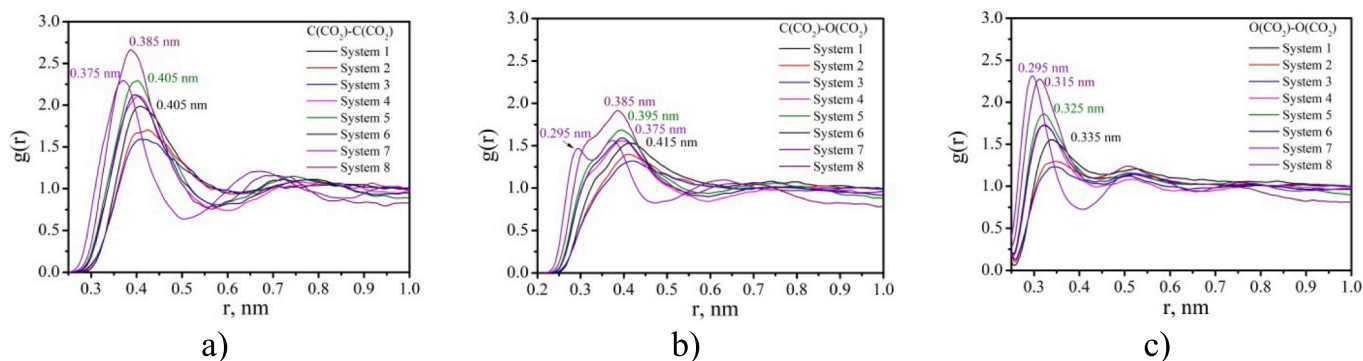


Fig. 9. Radial distribution functions between carbon dioxide atoms.

highest density. In this system, the pre-peak evolves into a well-defined, distinct peak with a maximum at 0.295 nm. The appearance of this peak indicates increased ordering within the nearest solvation shell and a higher proportion of CO₂ molecules in close contact with each other. This finding qualitatively agrees with existing literature, which reports clearer boundaries of the first coordination sphere and enhanced correlations in scCO₂ as pressure (and density) increase [51,52].

Fig. 10 reveals distinct changes in the intensity and position of bands in the regions of O—H and N—H stretching vibrations (3175–3700 cm^{−1}), as well as in the C=O stretching vibration region (1630–1710 cm^{−1}). First, a general thermal effect is evident (Fig. 10a): the intensity of all characteristic bands increases at higher temperature (in particular, for MFA in pure scCO₂ at 473 K compared to 328 K). This is attributed to the increased amplitude of atomic thermal motion at elevated temperatures, which naturally leads to a higher intensity of the corresponding bands in the VDOS spectrum. Against this background, more specific changes occur both with variations in the conformational state of MFA (Systems 2 and 7) and with changes in system composition (Systems 2 and 5, 4 and 7). Let us examine in detail the factors responsible for the

shifts or intensity changes of each band.

In Systems 2 and 7 (Fig. 10d), MFA adopts different conformational states: in System 2 (pure scCO₂ at *T* = 433 K), the C2 conformer is present, whereas in System 7 (scCO₂ with added cellobiose at *T* = 433 K), the C1 conformer is observed. Although System 7 contains two cellobiose molecules, a comparative analysis of the stretching frequencies in the —NH and —COOH groups is possible because the MFA in System 7 is oriented such that the carboxyl group faces the scCO₂ phase and does not interact with cellobiose.

As shown in Fig. 10d, the C1 conformer, which features a strong intramolecular HB between the carbonyl oxygen and the hydrogen atom of the —NH group, exhibits an intense band at 3623 cm^{−1}, corresponding to O—H stretching vibrations, along with a broad, low-intensity band in the 3250–3500 cm^{−1} region, corresponding to N—H stretching vibrations. Upon transitioning to the C2 conformer (System 2), where the strong intramolecular HB between the —NH hydrogen and the carbonyl oxygen is absent, a distinct band appears at 3504 cm^{−1}, corresponding to N—H stretching vibrations. Previous studies have also shown that the transition from polymorph 1 to polymorph 2 results in a

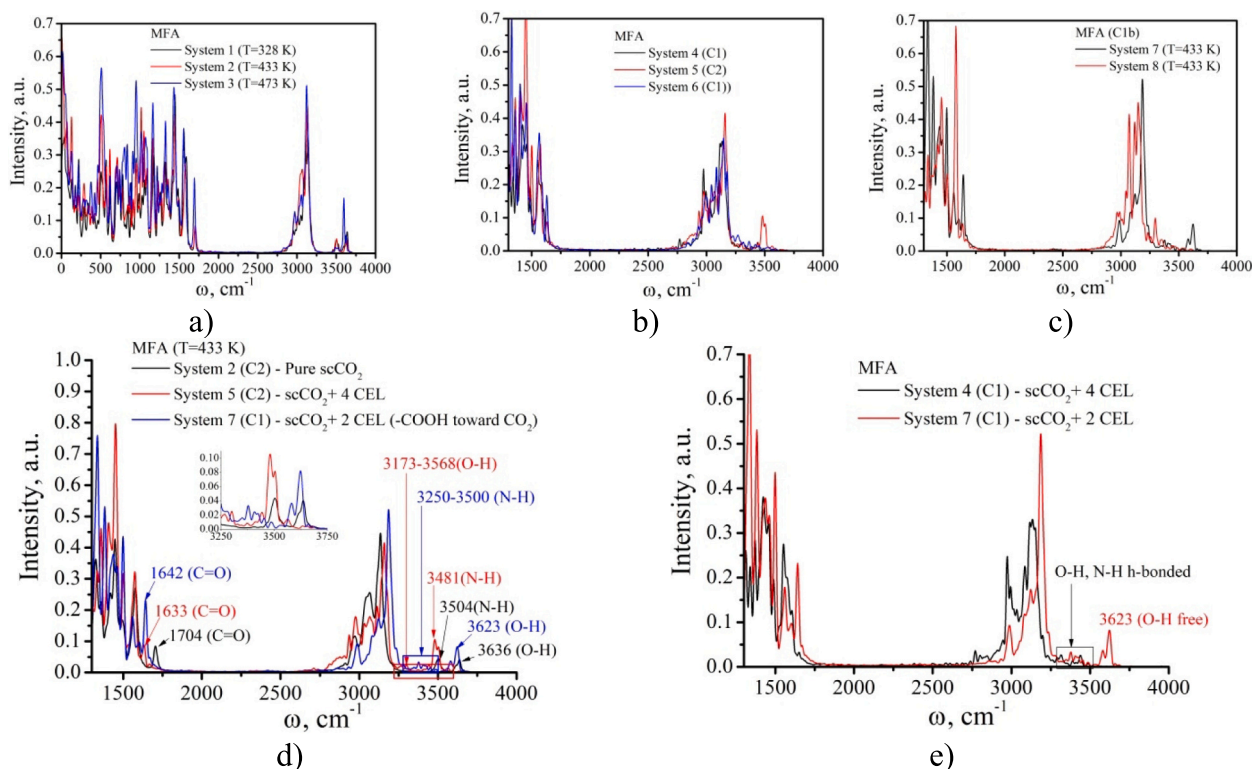


Fig. 10. Vibrational density of states spectra of mefenamic acid in various environments, derived from *ab initio* molecular dynamics simulations.

band shift from 3312 cm^{-1} to 3353 cm^{-1} , indicating a weakening of the intramolecular HB involving the —NH group [53,54]. Additionally, a red shift in the C=O stretching vibration frequency from 1704 cm^{-1} to 1642 cm^{-1} is observed upon transitioning from the C2 to the C1 conformer. This shift is attributed to the strong intramolecular HB present in the C1 conformer and is in good agreement with the data reported by Oparin et al. [12].

In addition to the conformational state of MFA, its interaction with cellobiose also affects the intensity and position of the N—H and O—H stretching bands. By comparing Systems 2 and 5, where the C2 conformer is present, as well as Systems 4 and 7, where MFA adopts the C1 conformation, a significant red shift (Fig. 10d) and a decrease in the intensity of the O—H stretching band are observed. These changes are attributed to the formation of HBs between the hydrogen atom of the

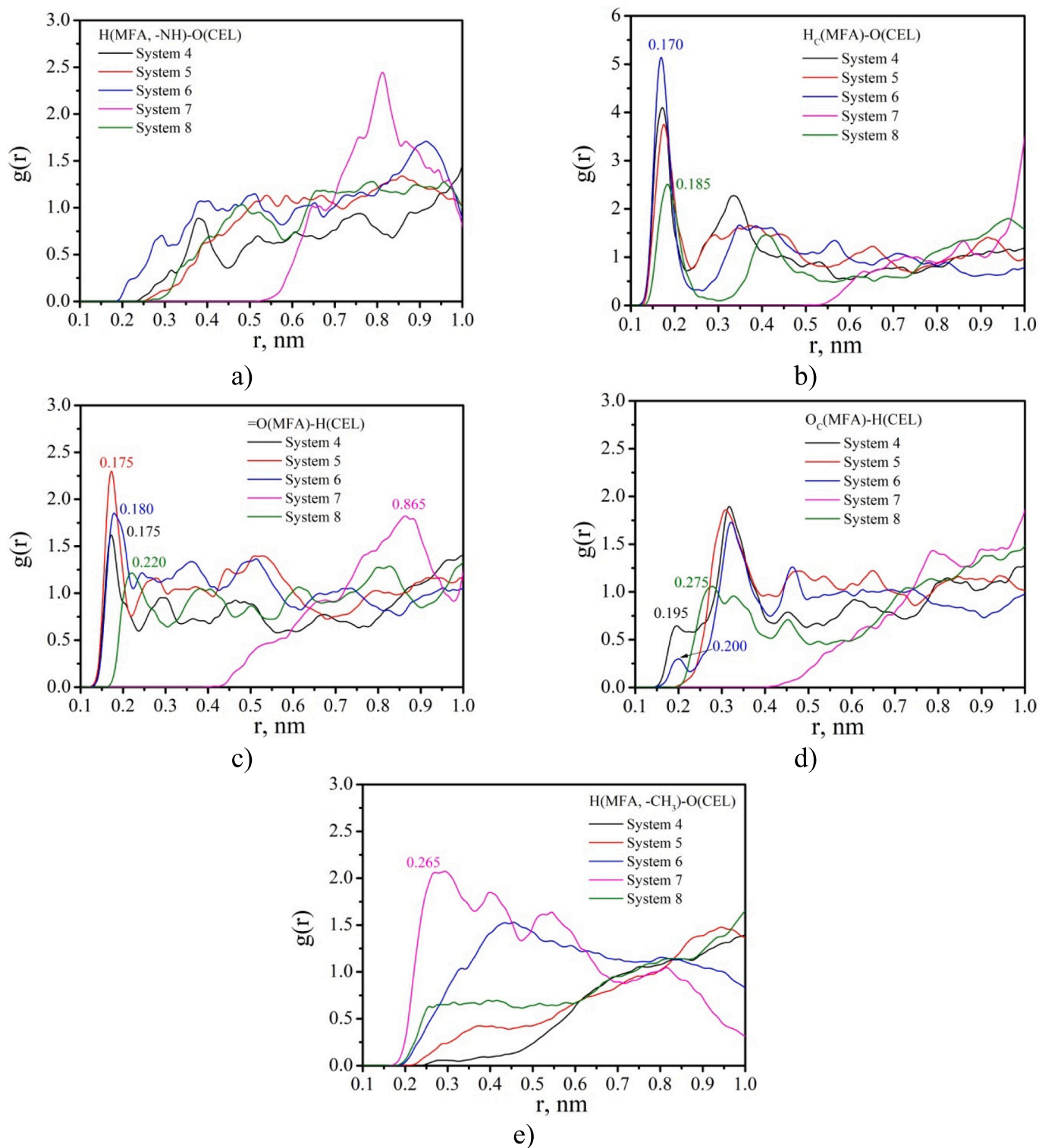


Fig. 11. Radial distribution functions between atoms of the hydroxyl groups of cellobiose (hydrogen and oxygen) and atoms of the methyl, carboxyl, and amino groups of MFA (hydrogen and oxygen).

carboxyl group and the hydroxyl groups of cellobiose. Additionally, a shift of the N—H stretching band from 3504 cm⁻¹ to 3481 cm⁻¹ (System 2 → System 5) is noted. This shift likely results from a cooperative effect in hydrogen bond formation: the hydrogen atom of the —NH group forms a hydrogen bond with the carbonyl oxygen atom (O_C) of the carboxyl group, while the carboxyl hydrogen simultaneously forms a hydrogen bond with a hydroxyl group of cellobiose (Fig. 11b).

Geometric analysis reveals consistent trends that corroborate the hydrogen-bonding interactions identified by VDOS and RDFs. The elongation of key bonds within MFA serves as a structural signature of intermolecular HB formation. As detailed in Table 2, a systematic increase in the average N—H and O—H bond lengths is observed specifically in systems where MFA forms HBs with cellobiose (Systems 4, 5, 6, and 8), compared to systems without such interactions (Systems 1–3 and 7). For instance, while the changes in absolute values (e.g., N—H from ~1.02 Å to ~1.03 Å) are within the range of thermal fluctuations, their consistent directional trend (bond elongation upon HB formation) is chemically meaningful and aligns perfectly with the predictions of VDOS.

This structural perturbation is most evident for the carboxyl O—H bond. In Systems 4–6 and 8, where a strong HB between the MFA carboxyl hydrogen and cellobiose is confirmed by sharp first peaks in the H_C(MFA)-O(CEL) RDFs (Fig. 11b), the O—H bond is consistently longer (~1.00–1.01 Å). This contrasts clearly with System 7, where despite the presence of cellobiose, the carboxyl group faces the scCO₂ phase and forms no HBs with the surface, thus retaining its shorter “free” bond length of ~0.99 Å. This clear distinction provides quantitative geometric evidence that directly mirrors the VDOS data: the red-shifted O—H stretching bands in the VDOS spectra (Figs. 10d,e) correspond precisely to the systems with elongated O—H bonds in the structural analysis.

Table 3 presents the average number of HBs formed between MFA and the hydroxyl groups of cellobiose. It can be observed that when MFA adopts the C1 conformational state (Systems 4, 6, and 8), most HBs are formed via the carboxyl hydrogen atom, while the carboxyl oxygen atoms are less frequently involved in intermolecular interactions with specific sites on cellobiose. This observation is corroborated by the analysis of RDFs (Fig. 11c, d), which show that the first peaks at distances corresponding to the geometric criteria for HB formation are lower in intensity than those for the RDF of H_C(MFA)-O(CEL). It is also worth noting that the probability of HB formation via the O_C atom (Fig. 11d) exists only for MFA in Systems 4 and 6; for the other systems, the first peak occurs at a distance greater than 0.26 nm. Conversely, in the C2 conformer, the oxygen atoms of the carboxyl group – particularly the carbonyl oxygen – are more actively involved in interactions with cellobiose, as further evidenced by the fact that the peak in the RDF for =O(MFA)-H(CEL) has the highest intensity (Fig. 11c). In this conformation, the carbonyl oxygen is sufficiently exposed to form two simultaneous HBs. The hydrogen bonding patterns observed between MFA and cellobiose align with recent classical MD simulations of MFA-impregnated NCC-aerogels [34].

To identify specific interactions between MFA and scCO₂, RDFs were analyzed for the oxygen and carbon atoms of CO₂ molecules relative to key atoms of MFA: the carboxyl hydrogen (H_C), amide hydrogen (H_N), methyl group hydrogens, and the carbonyl oxygen (=O). Although the

Table 3

Average number of hydrogen bonds between mefenamic acid and cellobiose.

System	Conf.	H _C	=O, O _C	Total
4	C1	0.68	0.61	1.30
5	C2	0.65	0.77	1.42
6	C1	0.83	0.75	1.58
7	C1	0	0	0
8	C1	0.55	0	0.55

features of these RDFs were less pronounced than in the MFA-cellobiose case, they indicate the presence of directed contacts between MFA and the solvent. The most significant feature was observed in the H_C(MFA)-O(CO₂) RDF for Systems 1 and 7 (Fig. 12). These systems exhibit a distinct peak at 0.235 nm, a distance that satisfies the geometric criterion for HB formation, suggesting the establishment of a weak HB between the carboxyl hydrogen of MFA and an oxygen atom of CO₂.

In Systems 4–6, the situation changes dramatically. The probability of finding a CO₂ oxygen atom near the MFA carboxyl hydrogen within a distance of less than 0.26 nm becomes negligible. This directly indicates that this hydrogen-bonding site is blocked and inaccessible to solvent interaction because it is engaged in a stronger interaction with cellobiose. This result is in accordance with the explanation for the pathway selectivity reported in [34], where the formation of stable HBs between the MFA carboxyl group and the aerogel surface was shown to block CO₂ attack, directing reactivity exclusively to the amino group.

Interactions of CO₂ with other sites on MFA are weaker. The probability of finding a CO₂ oxygen atom within 0.26 nm of the amide (H_N) or methyl group hydrogens is significantly lower than that for the carboxyl hydrogen (Fig. 12b and c). This confirms that the primary and most specific interaction site between MFA and CO₂ is the carboxyl hydrogen atom.

The analysis of the RDF between the carbonyl oxygen atom of MFA (=O) and the carbon atom of CO₂ (=O(MFA)-C(CO₂)) provides evidence of a LA-LB interaction (Fig. 12d). This is indicated by the presence of a first peak at short distances (up to approximately 0.35 nm), characteristic of directed non-covalent contacts. The strength of this interaction varies significantly across different systems. The highest contact probability is observed in Systems 1 and 7. In System 1 (a binary MFA/scCO₂ mixture at moderate temperature, 328 K), the solvent has direct access to the carbonyl group. In System 7, an additional factor enhancing the interaction is the high local density of CO₂ near the MFA carboxyl group, which remains unbound to cellobiose. The lowest probability is recorded in System 6, where the carboxyl group of MFA forms the greatest number of HBs with cellobiose. The RDF = O(MFA)-C(CO₂) demonstrates that, with such strong binding to cellobiose, the probability of the MFA carbonyl oxygen approaching CO₂ carbon atoms becomes negligible.

Therefore, it can be concluded that the carbonyl oxygen of MFA exhibits Lewis base properties only when the molecule's carboxyl group is not involved in strong, specific interactions with cellobiose. The formation of HBs between the carboxyl group and cellobiose likely alters the electron density or sterically hinders the carbonyl oxygen, completely suppressing its ability to engage in weak donor-acceptor

Table 2

Geometrical parameters of the mefenamic acid molecule for the studied systems.

System	Conf.	r(N—H), Å	r(O—H), Å	r(C=O), Å	∠H—O—C, degree	∠O—C—O, degree
1	C2	1.02 ± 0.03	0.99 ± 0.02	1.23 ± 0.03	106 ± 5	120 ± 3
2	C2	1.02 ± 0.03	0.99 ± 0.02	1.23 ± 0.03	106 ± 5	120 ± 3
3	C2	1.02 ± 0.03	0.99 ± 0.03	1.23 ± 0.03	106 ± 6	120 ± 4
4	C1	1.03 ± 0.03	1.01 ± 0.04	1.25 ± 0.03	109 ± 6	121 ± 3
5	C2	1.03 ± 0.03	1.01 ± 0.04	1.24 ± 0.02	109 ± 6	122 ± 4
6	C1	1.03 ± 0.03	1.01 ± 0.04	1.25 ± 0.03	109 ± 6	122 ± 3
7	C1	1.03 ± 0.04	0.99 ± 0.03	1.24 ± 0.02	107 ± 5	121 ± 3
8	C1	1.03 ± 0.03	1.00 ± 0.03	1.24 ± 0.03	108 ± 5	121 ± 3

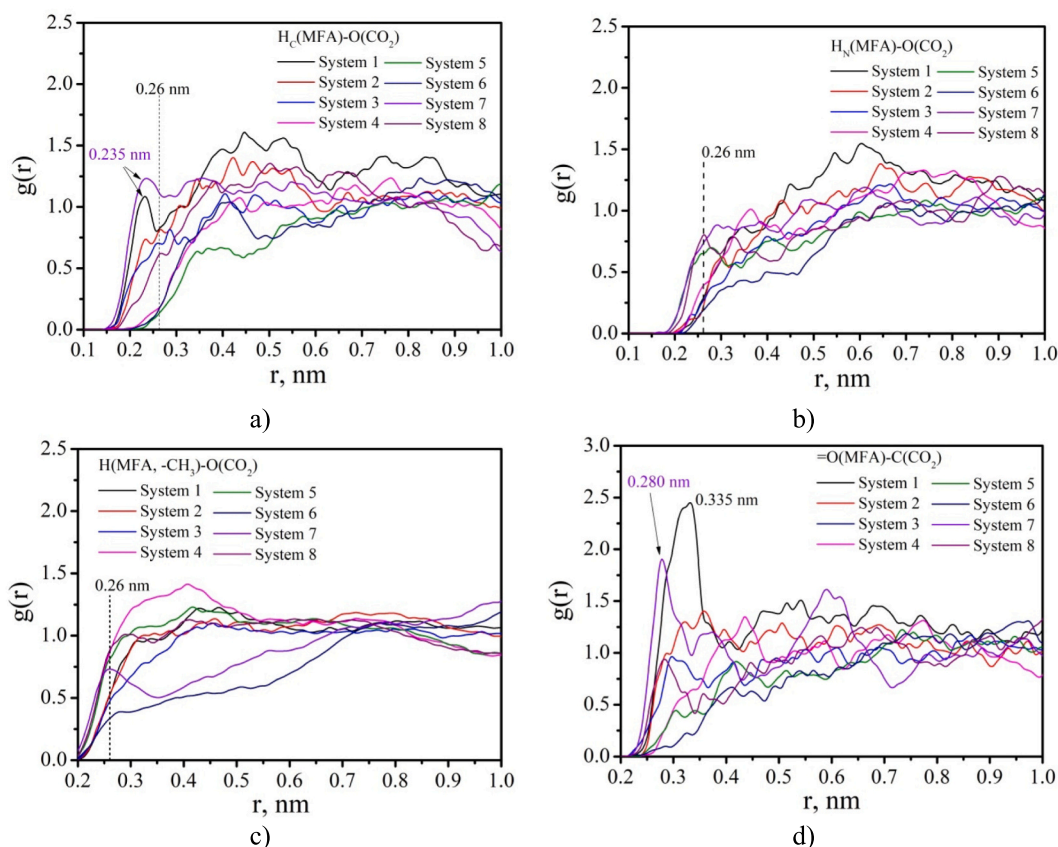


Fig. 12. Radial distribution functions between carbon/oxygen atoms of carbon dioxide and hydrogen/oxygen atoms of the methyl, carboxyl, and amino groups in MFA.

interactions with CO₂.

4. Conclusions

The comprehensive computational simulation conducted in this study, which combined classical MD-based solvation free energy calculations with detailed *ab initio* MD simulations, has provided an in-depth understanding of mefenamic acid behavior in scCO₂ and under conditions simulating its interaction with a cellulose-based carrier.

Thermodynamic analysis revealed a strong and non-monotonic dependence of MFA conformer solvation on the state parameters of scCO₂. At temperatures of 333 K and 403 K, increasing the density significantly enhances solvation, making ΔG_{solv} more negative. In contrast, at the elevated temperature of 483 K, an inverse trend is observed, where ΔG_{solv} becomes less negative upon transitioning to higher density, indicating the growing dominance of the entropic contribution. The maximum conformational selectivity of the medium – defined as a difference in ΔG_{solv} of up to approximately 3.7 kJ/mol between conformers – is observed specifically at high density and low temperature. Across the entire studied parameter range, the C2a conformer was identified as the most thermodynamically stable in the bulk scCO₂ phase, followed by the C2b conformer.

Ab initio MD simulations demonstrated that the presence of cellobiose, used as a cellulose model, fundamentally alters the conformational preferences of MFA compared to a pure scCO₂ environment. Specifically, there is a stabilization of type-b conformers (C1b, C2b), and the rotational freedom of both the carboxyl and aromatic groups is significantly restricted.

Notably, our study provides a clear rationale for a key experimental observation. NMR and IR spectroscopy fail to distinguish between conformers a and b of MFA in scCO₂, both in the bulk phase and under

aerogel confinement, as their signals appear coalesced. Our *ab initio* MD simulations directly reveal the underlying cause: the interconversion between these states is an ultrafast process occurring on a picosecond timescale, which we successfully captured within our 24-ps trajectories. Thus, the simulations explain the spectroscopic averaging by linking it to the rapid conformational dynamics at the molecular level.

The key atomistic mechanism for MFA – cellobiose interaction has been established: the formation of directional HBs between the functional groups of MFA and the hydroxyl groups of cellobiose. For the C1 conformer, binding primarily occurs via the carboxyl hydrogen atom, whereas for the C2 conformer, the carbonyl oxygen actively participates in HB formation. This interaction is directly reflected in the VDOS spectra as a red shift in the O–H and N–H stretching bands.

Therefore, the results of this study demonstrate that the efficient impregnation of MFA into a cellulose matrix within scCO₂ medium results from the synergy of two factors: 1) the thermodynamic predisposition for the existence of specific conformers (C2a, C2b) in the supercritical fluid, and 2) their subsequent specific and strong interaction with the carrier via hydrogen bonding. These insights provide a foundation for optimizing the loading process by adjusting pressure and temperature and may guide the future design of tailored carriers. The developed MFA-cellobiose interaction model facilitates the screening of other APIs, considering their conformational features. Future research should focus on simulating more extended cellulose surfaces and multilayer sorption.

CRediT authorship contribution statement

D.L. Gurina: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **K.A. Fashchevsky:** Visualization, Investigation. **Yu.A. Budkov:** Writing – review & editing. **M.G. Kiselev:**

Writing – review & editing, Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2026.129401>.

Data availability

No data was used for the research described in the article.

References

- [1] Sajal, N. H. (2025). Supercritical CO₂ as a green solvent: a comprehensive review of its applications in industrial processes. Available at SSRN 5225504. doi:<https://doi.org/10.2139/ssrn.5225504>.
- [2] A. Sinha, A. Bharti, P. Munshi, A review on solubility studies of industrially important compounds in supercritical CO₂, *ChemistrySelect* 10 (38) (2025) e02648, <https://doi.org/10.1002/slct.202502648>.
- [3] N.D. Machado, J.E. Mosquera, R.E. Martini, M.L. Goñi, N.A. Gañán, Supercritical CO₂-assisted impregnation/deposition of polymeric materials with pharmaceutical, nutraceutical, and biomedical applications: a review (2015–2021), *J. Supercrit. Fluids* 191 (2022) 105763, <https://doi.org/10.1016/j.supflu.2022.105763>.
- [4] Z. Liu, S. Zhang, B. He, S. Wang, F. Kong, Synthesis of cellulose aerogels as promising carriers for drug delivery: a review, *Cellulose* 28 (5) (2021) 2697–2714, <https://doi.org/10.1007/s10570-021-03734-9>.
- [5] L.Y. Long, Y.X. Weng, Y.Z. Wang, Cellulose aerogels: synthesis, applications, and prospects, *Polymers* 10 (6) (2018) 623, <https://doi.org/10.3390/polym10060623>.
- [6] J. Krivokapić, J. Ivanović, M. Krkobabić, J. Arsenijević, S. Ibrić, Supercritical fluid impregnation of microcrystalline cellulose derived from the agricultural waste with ibuprofen, *Sustain. Chem. Pharm.* 21 (2021) 100447, <https://doi.org/10.1016/j.scp.2021.100447>.
- [7] J.M. Lopes, A.N. Mustapa, M. Pantić, M.D. Bermejo, Á. Martín, Z. Novak, M. J. Cocero, Preparation of cellulose aerogels from ionic liquid solutions for supercritical impregnation of phytol, *J. Supercrit. Fluids* 130 (2017) 17–22, <https://doi.org/10.1016/j.supflu.2017.07.018>.
- [8] C. Darpentigny, P.R. Marcoux, M. Menneteau, B. Michel, F. Ricoul, B. Jean, G. Nonglaton, Antimicrobial cellulose nanofibril porous materials obtained by supercritical impregnation of thymol, *ACS Appl. Bio Mater.* 3 (5) (2020) 2965–2975, <https://doi.org/10.1021/acsbm.0c00033>.
- [9] Grout, S., Buwalda, S., & Budtova, T. (2022). Tuning bio-aerogel properties for controlling drug delivery. Part 2: Cellulose-pectin composite aerogels. *Biomater. Adv.*, 135, 212732. <https://doi.org/https://doi.org/10.1016/j.bioadv.2022.212732>.
- [10] A.N. Mustapa, A. Martín, L.M. Sanz-Moral, M. Rueda, M.J. Cocero, Impregnation of medicinal plant phytochemical compounds into silica and alginate aerogels, *J. Supercrit. Fluids* 116 (2016) 251–263, <https://doi.org/10.1016/j.supflu.2016.06.002>.
- [11] A.Z. Hezave, M.H. Khademi, F. Esmailzadeh, Measurement and modeling of mefenamic acid solubility in supercritical carbon dioxide, *Fluid Phase Equilibria* 313 (2012) 140–147, <https://doi.org/10.1016/j.fluid.2011.09.031>.
- [12] R.D. Oparin, M.A. Krestyaninov, D.V. Ivlev, M.G. Kiselev, Molecular mechanism of conformational crossover of mefenamic acid molecules in scCO₂, *Materials* 16 (4) (2023) 1403, <https://doi.org/10.3390/ma16041403>.
- [13] P.V. Klimovich, D.L. Mobley, Predicting hydration free energies using all-atom molecular dynamics simulations and multiple starting conformations, *J. Comput. Aided Mol. Des.* 24 (4) (2010) 307–316, <https://doi.org/10.1007/s10822-010-9343-7>.
- [14] S. Mittal, S. Prasad, R.S. Payal, S. Ravi, Ab-initio study on the covalent nature of hydrogen bonding in cellobiose, *Cellulose* 30 (2) (2023) 689–704, <https://doi.org/10.1007/s10570-022-04939-2>.
- [15] N. Barrios, J.G. Parra, P. Iza, R.A. Venditti, L. Pal, Density functional theory study of the cellobiose model system for unveiling cellulose–water interactions to enhance dewatering and drying, *J. Phys. Chem. A* (2025), <https://doi.org/10.1021/acs.jpca.5c01596>.
- [16] M. Saharay, S. Balasubramanian, Ab initio molecular-dynamics study of supercritical carbon dioxide, *J. Chem. Phys.* 120 (20) (2004) 9694–9702, <https://doi.org/10.1063/1.1701838>.
- [17] W. Mi, P. Ramos, J. Maranhao, M. Pavanello, Ab initio structure and dynamics of CO₂ at supercritical conditions, *J. Phys. Chem. Lett.* 10 (24) (2019) 7554–7559, <https://doi.org/10.1021/acs.jpclett.9b03054>.
- [18] A.G. Reveco-Chilla, L.M. Valenzuela, J.M. del Valle, E.J. Maginn, Use of molecular dynamics simulations to estimate the solubility of menadione in supercritical CO₂ using Chrastil's model, *Fluid Phase Equilib.* 433 (2017) 112–118, <https://doi.org/10.1016/j.fluid.2016.10.017>.
- [19] V. Reddy, M. Saharay, Solubility of caffeine in supercritical CO₂: a molecular dynamics simulation study, *J. Phys. Chem. B* 123 (45) (2019) 9685–9691, <https://doi.org/10.1021/acs.jpcc.9b08351>.
- [20] Y. Qin, X. Yang, Y. Zhu, J. Ping, Molecular dynamics simulation of interaction between supercritical CO₂ fluid and modified silica surfaces, *J. Phys. Chem. C* 112 (33) (2008) 12815–12824, <https://doi.org/10.1021/jp711964e>.
- [21] D.L. Gurina, Y.A. Budkov, M.G. Kiselev, Impregnation of poly (methyl methacrylate) with carbamazepine in supercritical carbon dioxide: molecular dynamics simulation, *J. Phys. Chem. B* 124 (38) (2020) 8410–8417, <https://doi.org/10.1021/acs.jpcc.0c05657>.
- [22] D.L. Gurina, Y.A. Budkov, M.G. Kiselev, A molecular insight into poly (methyl methacrylate) impregnation with mefenamic acid in supercritical carbon dioxide: a computational simulation, *J. Mol. Liq.* 337 (2021) 116424, <https://doi.org/10.1016/j.molliq.2021.116424>.
- [23] D.L. Gurina, Y.A. Budkov, M.G. Kiselev, Polylactide nanoparticle impregnation with carbamazepine in supercritical media and its subsequent release in liquid solvents: insights from molecular simulation, *J. Mol. Liq.* 352 (2022) 118758, <https://doi.org/10.1016/j.molliq.2022.118758>.
- [24] M.S. Lee, B. Peter McGrail, R. Rousseau, V.A. Glezakou, Structure, dynamics and stability of water/scCO₂/mineral interfaces from ab initio molecular dynamics simulations, *Sci. Rep.* 5 (1) (2015) 14857, <https://doi.org/10.1038/srep14857>.
- [25] M.J. Abraham, T. Murtola, R. Schulz, S. Páll, J.C. Smith, B. Hess, E. Lindahl, GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers, *SoftwareX* 1 (2015) 19–25, <https://doi.org/10.1016/j.softx.2015.06.001>.
- [26] M. Parrinello, A. Rahman, Polymorphic transitions in single crystals: a new molecular dynamics method, *J. Appl. Phys.* 52 (12) (1981) 7182–7190, <https://doi.org/10.1063/1.328693>.
- [27] W.L. Jorgensen, D.S. Maxwell, J. Tirado-Rives, Development and testing of the OPLS all-atom force field on conformational energetics and properties of organic liquids, *J. Am. Chem. Soc.* 118 (45) (1996) 11225–11236, <https://doi.org/10.1021/ja9621760>.
- [28] Z.H.U. Aimei, X. Zhang, L.I.U. Qinglin, Q. Zhang, A fully flexible potential model for carbon dioxide, *Chin. J. Chem. Eng.* 17 (2) (2009) 268–272, [https://doi.org/10.1016/S1004-9541\(08\)60204-9](https://doi.org/10.1016/S1004-9541(08)60204-9).
- [29] W.L. Jorgensen, J. Tirado-Rives, Potential energy functions for atomic-level simulations of water and organic and biomolecular systems, *Proc. Natl. Acad. Sci.* 102 (19) (2005) 6665–6670, <https://doi.org/10.1073/pnas.0408037102>.
- [30] L.S. Dodda, J.Z. Vilseck, J. Tirado-Rives, W.L. Jorgensen, 1.14* CM1A-LBCC: localized bond-charge corrected CM1A charges for condensed-phase simulations, *J. Phys. Chem. B* 121 (15) (2017) 3864–3870, <https://doi.org/10.1021/acs.jpcc.7b00272>.
- [31] L.S. Dodda, I. Cabeza de Vaca, J. Tirado-Rives, W.L. Jorgensen, LigParGen web server: an automatic OPLS-AA parameter generator for organic ligands, *Nucleic Acids Res.* 45 (W1) (2017) W331–W336, <https://doi.org/10.1093/nar/gkx312>.
- [32] Noel M. O'Boyle, Michael Banck, Craig A. James, Chris Morley, Tim Vandermeersch, Geoffrey R. Hutchison, Open Babel: an open chemical toolbox, *J. Cheminform.* 3 (2011) 33, <https://doi.org/10.1186/1758-2946-3-33>.
- [33] P.J. L. (2003). NIST standard reference database number 69. *NIST Chemistry WebBook*.
- [34] R.D. Oparin, M.A. Krestyaninov, D.V. Ivlev, A.A. Dyshin, M.S. Kuz'mikov, M. G. Kiselev, Reaction mechanism between carbon dioxide and mefenamic acid in saturated solution and within the confinement of a nanocrystalline cellulose aerogel, *J. Mol. Liq.* (2025) 128146, <https://doi.org/10.1016/j.molliq.2025.128146>.
- [35] T.D. Kühne, M. Iannuzzi, M. Del Ben, V.V. Rybkin, P. Seewald, F. Stein, J. Hutter, CP2K: an electronic structure and molecular dynamics software package-quickstep: efficient and accurate electronic structure calculations, *J. Chem. Phys.* 152 (19) (2020), <https://doi.org/10.1063/5.0007045>.
- [36] S. Goedecker, M. Teter, J. Hutter, Separable dual-space Gaussian pseudopotentials, *Phys. Rev. B* 54 (3) (1996) 1703, <https://doi.org/10.1103/PhysRevB.54.1703>.
- [37] C. Hartwigsen, S. Goedecker, J. Hutter, Relativistic separable dual-space Gaussian pseudopotentials from H to Rn, *Phys. Rev. B* 58 (7) (1998) 3641, <https://doi.org/10.1103/PhysRevB.58.3641>.
- [38] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu, *J. Chem. Phys.* 132 (15) (2010), <https://doi.org/10.1063/1.3382344>.
- [39] S. Grimme, S. Ehrlich, L. Goerigk, Effect of the damping function in dispersion corrected density functional theory, *J. Comput. Chem.* 32 (7) (2011) 1456–1465, <https://doi.org/10.1002/jcc.21759>.
- [40] L. Martínez, R. Andrade, E.G. Birgin, J.M. Martínez, PACKMOL: a package for building initial configurations for molecular dynamics simulations, *J. Comput. Chem.* 30 (13) (2009) 2157–2164, <https://doi.org/10.1002/jcc.21224>.
- [41] W. Humphrey, A. Dalke, K. Schulten, VMD: visual molecular dynamics, *J. Mol. Graph.* 14 (1) (1996) 33–38, [https://doi.org/10.1016/0263-7855\(96\)00018-5](https://doi.org/10.1016/0263-7855(96)00018-5).

- [42] M. Brehm, M. Thomas, S. Gehrke, B. Kirchner, TRAVIS—a free analyzer for trajectories from molecular simulation, *J. Chem. Phys.* 152 (16) (2020), <https://doi.org/10.1063/5.0005078>.
- [43] M. Brehm, B. Kirchner, TRAVIS-a free analyzer and visualizer for Monte Carlo and molecular dynamics trajectories, *J. Chem. Inf. Model.* (2011), <https://doi.org/10.1021/ci200217w>.
- [44] I. Khodov, V. Sobornova, V. Mulloyarova, K. Belov, A. Dyshin, L.B. de Carvalho, M. Kiselev, Exploring the conformational equilibrium of mefenamic acid released from silica aerogels via NMR analysis, *Int. J. Mol. Sci.* 24 (8) (2023) 6882, <https://doi.org/10.3390/ijms24086882>.
- [45] K.V. Belov, L.A. Batista de Carvalho, A.A. Dyshin, S.V. Efimov, I.A. Khodov, The role of hidden conformers in determination of conformational preferences of mefenamic acid by NOESY spectroscopy, *Pharmaceutics* 14 (11) (2022) 2276, <https://doi.org/10.3390/pharmaceutics14112276>.
- [46] Y. Garrabos, V. Chandrasekharan, M.A. Echargui, F. Marsault-Herail, Density effect on the Raman fermi resonance in the fluid phases of CO₂, *Chem. Phys. Lett.* 160 (3) (1989) 250–256, [https://doi.org/10.1016/0009-2614\(89\)87591-8](https://doi.org/10.1016/0009-2614(89)87591-8).
- [47] Y. Yagi, H. Tsugane, H. Inomata, S. Saito, Density dependence of fermi resonance of supercritical carbon dioxide, *J. Supercrit. Fluids* 6 (3) (1993) 139–142, [https://doi.org/10.1016/0896-8446\(93\)90012-M](https://doi.org/10.1016/0896-8446(93)90012-M).
- [48] M. Poliakkoff, S.M. Howdle, S.G. Kazarian, Vibrational spectroscopy in supercritical fluids: from analysis and hydrogen bonding to polymers and synthesis, *Angew. Chem. Int. Ed. Engl.* 34 (12) (1995) 1275–1295, <https://doi.org/10.1002/ange.199512751>.
- [49] Y. Kachi, T. Tsukahara, Y. Kayaki, T. Ikariya, J. Sato, Y. Ikeda, Raman spectral shifts of CO₂ as measure of CO₂-philicity of solutes in supercritical carbon dioxide, *J. Supercrit. Fluids* 40 (1) (2007) 20–26, <https://doi.org/10.1016/j.supflu.2006.05.002>.
- [50] S. Balasubramanian, A. Kohlmeier, M.L. Klein, Ab initio molecular dynamics study of supercritical carbon dioxide including dispersion corrections, *J. Chem. Phys.* 131 (14) (2009), <https://doi.org/10.1063/1.3245962>.
- [51] M. Saharay, S. Balasubramanian, Evolution of intermolecular structure and dynamics in supercritical carbon dioxide with pressure: an ab initio molecular dynamics study, *J. Phys. Chem. B* 111 (2) (2007) 387–392, <https://doi.org/10.1021/jp065679t>.
- [52] A.K. Adya, C.J. Wormald, Intra and intermolecular structure in the condensed phases of ethylene, ethane and carbon dioxide by neutron diffraction, *Mol. Phys.* 77 (6) (1992) 1217–1246, <https://doi.org/10.1080/00268979200103091>.
- [53] N. Abbas, I.D. Oswald, C.R. Pulham, Accessing mefenamic acid form II through high-pressure recrystallisation, *Pharmaceutics* 9 (2) (2017) 16, <https://doi.org/10.3390/pharmaceutics9020016>.
- [54] R.K. Gilpin, W. Zhou, Infrared studies of the thermal conversion of mefenamic acid between polymorphic states, *Vib. Spectrosc.* 37 (1) (2005) 53–59, <https://doi.org/10.1016/j.vibspec.2004.06.001>.